

Node features ($N \times 4$)

variable v_i : $[\ell, 1, 0, 0]$

X-check c_j : $[s_j, 0, 1, 0]$

Z-check c_j : $[s_j, 0, 0, 1]$

FiLM variant only (not used in Sec. 5.3)

FiLM generator $\hat{p} \rightarrow \{(\gamma_\ell, \beta_\ell)\}_{\ell=1}^L$

Typed message passing $\times L$

Input proj.
Linear $\rightarrow$ LN $\rightarrow$ ReLU

$\mathbf{h}^{(0)}$

Edge MLP _{τ}
 $[\mathbf{h}_i || \mathbf{h}_j]$

$\Sigma_{j \in \mathcal{N}(i)}$

FiLM

$\gamma \odot \mathbf{m} + \beta$

GRU

$(\mathbf{m}_i, \mathbf{h}_i)$

Residual

+ LN

$\mathbf{h}^{(L)}$

Readout MLP
variable nodes

$\Lambda_i + \Delta_i$ to BP

Δ_i

$\ell = \ln \frac{1-p}{p}$; s_j : syndrome bit

repeat for each of the L layers